Problem set 4

PPHA 31102 Statistics for Data Analysis II: Regressions

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1. Trade Policy

Tariffs frequently impact the flow of goods internationally. In this section, you will draw on data from one of these tariffs, investigating how these disruptions impact exports. For the next five questions, you will use the data posted on Canvas as *trade_policy.csv*. This data contains several variables: post is a binary variable indicating the tariff has been introduced (post = 1, pre = 0); *treat* is a binary variable that equals 1 if a country is "impacted" by the tariff barrier (i.e., a regulation increasing the cost of exports) and 0 if a country is "not impacted"; *air_conditioners_pc* is the number of air conditioners exported per capita by a given country in a year.

1.1.

Use a difference-in-differences design to estimate the impact of the tariff on exports of air conditioners. Regress air_conditioners_pc on post, treat, and the interaction of post and treat. Which coefficient reveals the change in exports after the tariff was introduced? Explain and interpret the result.

Answer

```
tp$interaction <- tp$treat * tp$post
model <- lm(air_conditioners_pc ~ treat + post + interaction, data = tp)
summary(model)
```

```
##
## Call:
## lm(formula = air_conditioners_pc ~ treat + post + interaction,
##     data = tp)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -1.2503 -0.9606 -0.6648 -0.1450 19.5300
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.96390    0.06872  14.026  < 2e-16 ***
## treat       -0.22753    0.08966  -2.538  0.01120 *
## post         0.28641    0.09839   2.911  0.00363 **
## interaction -0.63417    0.12837  -4.940 8.19e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.809 on 3276 degrees of freedom
## Multiple R-squared:  0.02855,    Adjusted R-squared:  0.02766
## F-statistic: 32.09 on 3 and 3276 DF,  p-value: < 2.2e-16
```

The coefficient that reveals the change in exports after the tariff was introduced, is the coefficient on the interaction between *treat* and *post* ($treat * post$). In this case, the result of the interaction reveals that the treated group decreased, after the tariffs, by 0.63417 air conditioners per capita relative to the control group.

1.2.

A friend is suggesting that you should consider taking the log of your outcome, air_conditioners_pc. What are some potential concerns regarding this suggestion? Describe two.

Answer

To consider the log of the outcomes from `air_conditioners_pc` might be misleading because of the near-zero values present in the dataframe, that after log(), generate extreme values that potentially distort the regression. Moreover, taking the log changes the interpretation from units to percentage changes, and the friend does not provide any theoretical justification for this transformation. Interpreting results in air conditioners per capita feels more intuitive in this context.

1.3

The difference-in-differences design is valid when treatment and control have common trends. What is a common trend? Write down the definition. Next, give an example (about this case) when this assumption is likely violated.

Answer

A common trend, in the context of difference-in-differences, refers to the idea that in the absence of treatment, both groups would have experienced the same change in the outcome over time.

The idea of a common trend might be likely violated if, for instance, the treated countries experienced an unusual increase in air conditioner demand during the pre-treatment period that the control countries did not. In this case, the two groups would have been on different trajectories even without the tariff.

2. The Oregon Health Insurance Experiment

2.1.

Consider the variable ever_medicaid. Did everyone who won the lottery take up Medicaid? Calculate the mean of ever_medicaid among those who won the lottery. Are there some people who lost the lottery who take up Medicaid? Calculate the mean of ever_medicaid for those that lost the lottery. Is the causal effect of winning the lottery likely to be the same as the causal effect of actually receiving Medicaid?

Answer

```
mean(ohie$ever_medicaid[ohie$treated == 1])
```

```
## [1] 0.4255519
```

```
mean(ohie$ever_medicaid[ohie$treated == 0])
```

```
## [1] 0.1857241
```

Answering the first question, no. Only 42.5% of those who won the lottery had Medicaid after treatment. And about 18.5% of those who lost the lottery had Medicaid after treatment.

With respect to the casual effect of winning the lottery being the same as the causal effect of actually receiving Medicaid, it is unlikely. Since not all the winners enroll and some losers do, the causal effect of winning the lottery is potentially different from the causal effect of actually receiving Medicaid.

2.2.

Estimate the "First-stage" regression:

$$ever\_medicaid = \alpha 0 + \alpha 1 * treated + \alpha 2 * numhh_list + e$$

Interpret the regression coefficient on $\alpha 1$.

Answer

```
model <- lm(ever_medicaid ~ treated + numhh_list, data = ohie)
summary(model)
```

```
##
## Call:
## lm(formula = ever_medicaid ~ treated + numhh_list, data = ohie)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4394 -0.3914 -0.1951  0.5606  0.8529
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.243181   0.012650  19.224  < 2e-16 ***
## treated      0.244275   0.008133  30.033  < 2e-16 ***
## numhh_list  -0.048041   0.009380  -5.121 3.08e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4467 on 12226 degrees of freedom
## Multiple R-squared:  0.06897,    Adjusted R-squared:  0.06882
## F-statistic: 452.9 on 2 and 12226 DF,  p-value: < 2.2e-16
```

In this case, the coefficient on treated ($\alpha 1 * treated$) is 0.244, which means that winning the lottery increases the probability of having Medicaid by 24.4 percentage points. This suggests that winning the lottery has a significant positive effect on Medicaid enrollment.

2.3.

Is treated likely to be a valid instrument for ever_medicaid in terms of the three key assumptions: instrument relevance, independence assumption, and the exclusion restriction? If there is a specific statistical test that helps validate the plausibility of each assumption, please refer to this evidence

Answer

For each of the three assumptions:

1. Instrument relevance: Yes, we do have a valid and relevant instrument since the p-value is highly statistically significant ($<$ 2e-16), and the $t > 3.2$.
2. Independence assumption: Yes, the instrument is randomly assigned since is a lottery, so it should be independent of potential outcomes.
3. Exclusion restriction: Yes, the exclusion restriction is likely satisfied because winning the lottery should only affect the outcomes through its effect on Medicaid enrollment, and not through any other channels.

For validation of the instrument relevance through a test, we can consider an F-test for the first stage regression and making sure that F is large enough (>10) to avoid weak instruments problem. In the case of our data, the F-statistic is 452.9, which is significantly larger than 10.

The other assumptions don't have a specific statistical test, but it's more of a theoretical justification based on the design of the experiment (random assignment) and the nature of the instrument (winning a lottery).

2.4.

One possible way to estimate the effect of receiving health insurance is the ratio of the coefficient on the instrument in the reduced form model to the coefficient on the instrument in the first-stage regression. Let's again focus on the outcome count visit dr. Recall, the reduced form is the regression we've already been running in Problem Sets #1 and #3:

$$count\_visit\_dr = \beta_0 + \beta_1 treated + \beta_2 numhh_l ist + \epsilon$$

Calculate and report the ratio of the reduced form to the first stage. Note that technically, we should use the same number of observations in our first-stage and reduced form, so you should calculate the first stage excluding rows with missing values of count visit dr. Interpret your result

Answer

```
df <- ohie %>% filter(!is.na(count_visit_dr))

model_1 <- lm(ever_medicaid ~ treated + numhh_list, data = df)
coef(model_1)["treated"]
```

```
##  treated
## 0.243519
```

```
model_2 <- lm(count_visit_dr ~ treated + numhh_list, data = df)
coef(model_2)["treated"]
```

```
##   treated
## 0.5935102
```

Now, for ratio we need the coefficients from `treated` of both regressions. So for $B_{2step} = \frac{\rho}{\phi}$:

```
Beta_2step <- coef(model_2)["treated"] / coef(model_1)["treated"]
Beta_2step
```

```
##  treated
## 2.437223
```

So the ratio of the reduced form to the first stage is 2.437. This means that enrolling in Medicaid increases the number of doctor visits by approximately 2.437 visits.

2.5.

Let's next estimate the IV model via two-stage least squares (2SLS). Manually estimate the two stages:

(a) First-stage: Run the regression $ever\_medicaid = \alpha_0 + \alpha_1 treated + \alpha_2 numhh\_list + \epsilon$ **and compute prediceted values:** $ever\_\hat{medicaid} = \hat{\alpha_0} + \hat{\alpha_1} treated + \hat{\alpha_2} numhh\_list$

(b) Second-stage: Run the regression $count\_visit\_dr = \beta_0 + \beta_1 ever\_\hat{medicaid} + \beta_2 numhh\_list + \epsilon$

Again, you should only estimate the first stage on the subset of the data where count visit dr is not missing. How does your result compare to the previous estimate? You can now also conduct inference–is the IV estimate statistically significant?

Answer

```
# first stage lm
model_1sls <- lm(ever_medicaid ~ treated + numhh_list, data = df)
df$ever_medicaid_hat <- fitted(model_1sls)

# second stage lm
model_2sls <- lm(count_visit_dr ~ ever_medicaid_hat + numhh_list, data = df)
summary(model_2sls)
```

```
##
## Call:
## lm(formula = count_visit_dr ~ ever_medicaid_hat + numhh_list,
##     data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
##  -6.760  -5.760  -3.760  -0.028 137.834
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)         7.7098     0.4417  17.453  < 2e-16 ***
## ever_medicaid_hat   2.4372     0.8893   2.740  0.00614 **
## numhh_list         -2.0189     0.2487  -8.118 5.19e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.86 on 12155 degrees of freedom
## Multiple R-squared:  0.006276,   Adjusted R-squared:  0.006113
## F-statistic: 38.38 on 2 and 12155 DF,  p-value: < 2.2e-16
```

Based on the 2SLS results, the coefficient on `ever_medicaid_hat`is 2.437, which is the same as the ratio of the reduced form to the first stage that was calculated in Q2.4. The estimate is statistically significan at the 1% level (and $t = 2.740$). This result confirms that enrolling in Medicaid increases doctor visits by about 2.4 visits.

2.6.

Now, we will try including all the baseline characteristics as controls. However, unlike in Problem Set #3, let's account for a non-linearity in age by including a quadratic term, age^2.

Thus, the full-set of controls become:

$$numhh\_list, female, age, age^2, race\_white, hs_degree, college\_degree, health\_baseline$$

Re-estimate the 2SLS model including these controls and compare the size of the coefficient on $ever_medicaid$ and the standard error to the results in part (6).

Answer

```
df <- df %>% filter(!is.na(age) & !is.na(female) & !is.na(race_white) &
                     !is.na(hs_degree) & !is.na(college_degree) &
                     !is.na(health_baseline))

df$age2 <- df$age^2


first_stage <- lm(ever_medicaid ~ treated + numhh_list + female + age + age2 + race_whit

df$ever_medicaid_hat2 <- fitted(first_stage)

second_stage <- lm(count_visit_dr ~ ever_medicaid_hat2 + numhh_list + female + age + age

summary(second_stage)
```

```
##
## Call:
## lm(formula = count_visit_dr ~ ever_medicaid_hat2 + numhh_list +
##     female + age + age2 + race_white + hs_degree + college_degree +
##     health_baseline, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.446  -5.264  -3.140   0.394 140.101
```

```
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -1.9461222  1.4527176  -1.340 0.180387
## ever_medicaid_hat2  2.4246521  0.8715028   2.782 0.005408 **
## numhh_list         -1.5734505  0.2491363  -6.316 2.78e-10 ***
## female              2.4931789  0.2313475  10.777  < 2e-16 ***
## age                 0.2633161  0.0695231   3.787 0.000153 ***
## age2               -0.0030468  0.0008351  -3.649 0.000265 ***
## race_white          1.2373348  0.2376836   5.206 1.96e-07 ***
## hs_degree           1.0377172  0.2774203   3.741 0.000184 ***
## college_degree      2.2541510  0.4070297   5.538 3.12e-08 ***
## health_baseline     2.2871899  0.2584075   8.851  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.67 on 12115 degrees of freedom
## Multiple R-squared:  0.03367,    Adjusted R-squared:  0.03295
## F-statistic:  46.9 on 9 and 12115 DF,  p-value: < 2.2e-16
```

The coefficient on ever_medicaid with controls is 2.425, compared to 2.437 without controls in part 5. The point estimate is very similar and the standard error decreased slightly from 0.889 to 0.872, suggesting that adding controls improved precision slightly. The estimate remains statistically significant at the 1% level.

2.7.

For each endline outcome, run the IV regressions including controls using ivreg(). Fill in the table and discuss findings.

Answer

Endline outcome	(1) IV estimate	(2) S.E.
visit_dr	2.4246521	1.4395450
visit_er	0.0431894	0.1430542
out_of_pocket_spend	-283.03827	87.90313
health_score	-2.304e-03	3.961e-03
happy	2.651e-02	3.136e-02

Only `out_of_pocket_spending` showed a statistically significant effect, confirming that Medicaid enrollment reduces out-of-pocket spending by about $283. The other estimates are not statistically significant, so we can't rule out zero effects.

```r
ohie$age2 <- ohie$age^2

# visit_dr
iv_dr <- ivreg(count_visit_dr ~ ever_medicaid + numhh_list + female + age + age2 + race_

# visit_er
iv_er <- ivreg(count_visit_er ~ ever_medicaid + numhh_list + female + age + age2 + race_

# out_of_pocket_spend
iv_oop <- ivreg(out_of_pocket_spend ~ ever_medicaid + numhh_list + female + age + age2 +

# health_score
iv_hs <- ivreg(health_score ~ ever_medicaid + numhh_list + female + age + age2 + race_wh

# happy
iv_happy <- ivreg(happy ~ ever_medicaid + numhh_list + female + age + age2 + race_white
```